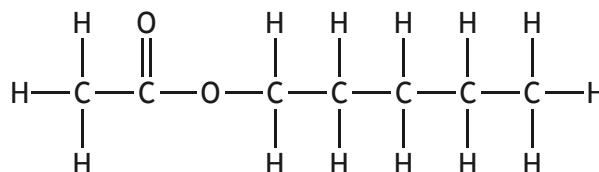


8. Sweets contain a wide variety of chemicals.

(a) Many sweets contain esters.

(i) The structure of an ester used to produce a pear flavour in some sweets is



Name this ester.

1

(ii) Name the type of reaction used to form esters.

1

(b) Fizzy sweets can contain citric acid, $\text{C}_6\text{H}_8\text{O}_7$, and sodium bicarbonate, NaHCO_3 .

When the sweets dissolve, the citric acid and sodium bicarbonate react together to make carbon dioxide gas.

(i) To calculate the mass of citric acid in a sweet, 5 sweets were dissolved in water and the resulting carbon dioxide was collected and measured.

Suggest why carbon dioxide can be collected over water.

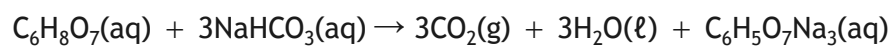
1



* X 8 1 3 7 6 0 1 2 2 *

8. (b) (continued)

- (ii) In one experiment, 5 sweets were dissolved, and 55 cm³ of carbon dioxide gas was produced.



GFM = 192 g

Calculate the mass of citric acid, in g, in one sweet.

2

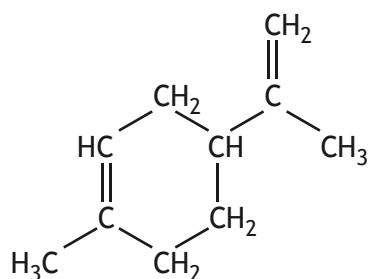
Take the volume of 1 mole of carbon dioxide to be 24 litres.



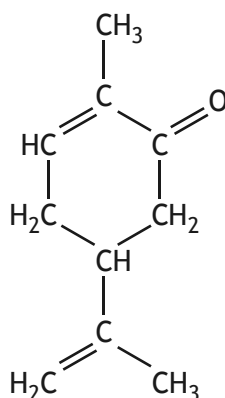
* X 8 1 3 7 6 0 1 2 3 *

8. (continued)

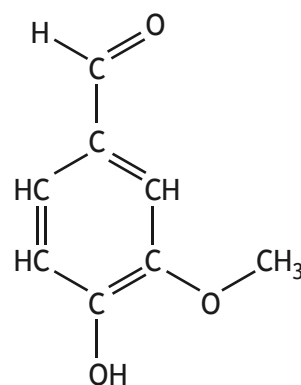
- (c) The distinctive smell of some sweets is due to molecules such as limonene, carvone and vanillin.



limonene



carvone



vanillin

- (i) Vanillin is an aldehyde and carvone is a ketone.
- (A) State the colour change that would be observed when aldehydes react with acidified potassium dichromate. 1
- (B) Suggest a different chemical that could be used to distinguish aldehydes from ketones. 1
- (ii) Limonene is a terpene consisting of joined isoprene units. State the number of isoprene units in a limonene molecule. 1

8. (c) (continued)

- (iii) 1 kg of natural vanillin costs £1050. To make a packet of sweets, 5 cm^3 of vanillin solution is used. This contains 0.184 g of vanillin per 100 cm^3 of solution.

Calculate the cost, in pence, of the natural vanillin required to make the packet of sweets.

2

[Turn over]



* X 8 1 3 7 6 0 1 2 5 *